YUSUPOV JAMSHIDBEK

Machine Learning Operation Engineer | Data Scientist

Email: jamusan@kookmin.ac.kr
Phone number: +998 91 001 27 23
GitHub: https://github.com/JamshidbekYusupov
LinkedIn: https://www.linkedin.com/in/jamshidbek-yusupov/

OBJECTIVE

MSc student in Business IT with hands-on experience building machine learning and deep learning models in Python and PyTorch and deploying them as containerized services with FastAPI and Docker. Seeking an MLOps and Data Scientist job to contribute to end-to-end model development, deployment, automated training and inference pipelines, and production monitoring.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, scikit-learn), SQL

Visualization: Matplotlib, Plotly Studio

Machine Learning: Supervised & Unsupervised Learning,Feature Engineering, Model Evaluation, Explainable AI (SHAP)

Deep Learning: PyTorch, CNN Image Classification, Video Classification, Transformer Architectures (Encoder-Decoder, Attention), OpenCV

Audio & Speech: Automatic Speech Recognition (ASR), Whisper Fine-tuning, Log-mel Spectrograms, Hugging Face Transformers/Datasets, WER Evaluation

MLOps & Deployment: FastAPI, Docker, CI/CD (GitHub Actions), Git, AWS (basics), Hugging Face Hub & Spaces, Weights & Biases (W&B), Gradio

Data & Storage: PostgreSQL, data pipelines, missing data handling, data validation

Mathematics: Linear Algebra, Probability, Statistics, Calculus

PROJECTS

Hotel Booking Cancellation Prediction — End-to-End ML Service | *Python, XGBoost, FastAPI, Docker*

- Built a complete ML pipeline from scratch: data cleaning, EDA, feature engineering, and model training (Logistic Regression, Random Forest, XGBoost) with hyperparameter tuning and SHAP-based explainability.
- Identified and removed target leakage in the raw data, replacing an unrealistically perfect baseline with an honest, generalizable model — documented the diagnosis and fix as part of the project.
- Served the final model through a FastAPI REST endpoint with input validation, then containerized the application with Docker for reproducible deployment.

Image Classification with PyTorch | *PyTorch, CNN, OpenCV*

- Implemented and trained convolutional neural networks in PyTorch for multi-class image classification, comparing a custom CNN with a transfer-learning baseline; evaluated with confusion matrices.
- Built the full training loop with data augmentation, learning-rate scheduling and early stopping; tracked accuracy and loss curves across experiments.

Video Classification with Deep Learning | *PyTorch, 2D CNN, OpenCV*

- Developed a video classification pipeline: frame extraction and preprocessing with OpenCV, followed by 2D CNN feature extraction and frame-level aggregation for clip-level prediction.
- Handled variable-length video data through frame sampling strategies and batched loading for efficient GPU training.

Uzbek Speech Recognition — Whisper Fine-tuning | *PyTorch, Hugging Face Transformers, Whisper, Gradio*

- Fine-tuning OpenAI Whisper-small for Uzbek automatic speech recognition, building the full pipeline from audio preprocessing (log-mel spectrograms) through training and WER evaluation using the Hugging Face Transformers ecosystem.
- Deployed the fine-tuned model as an interactive Gradio demo on Hugging Face Spaces for live testing and evaluation.

RESEARCH EXPERIENCE

Business IT Researcher | *Sep 2025 – Present*
KISLAB (Knowledge-based Intelligent Systems Lab), Kookmin University, South Korea

- Design and evaluate machine learning models for business analytics and decision support.
- Research techniques for handling missing data to improve predictive model performance and data integrity.

EDUCATION

MSc in Business IT | *Mar 2025 – Present*
Kookmin University, Seoul, South Korea

BSc in Economics and Management in Information Communication Technologies | *2020 – 2024*
Tashkent University of Information Technologies, Uzbekistan

Exchange Student | *2022 – 2023*

Shibaura Institute of Technology, Tokyo, Japan

LANGUAGES

Uzbek — Native | English — C1 (Fluent) | Russian — B1 (Intermediate) | Korean — A1 (Elementary)

AWARDS & SCHOLARSHIPS

- Full KOICA (Korea International Cooperation Agency) Scholarship — MSc in Business IT, Kookmin University, South Korea (2025 – Present).